SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.8: Commercial Arithmetic 1 — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.8: Commercial Arithmetic 1
Driving Question	DRIVING QUESTION: How can Amina — and every Kenyan — use commercial arithmetic to make smart, fair, and informed money decisions in markets, clubs, and across borders?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Amina's KSh 4,500 — Anchoring the Phenomenon & Building a Club Budget**	Students examined Amina's monthly income of KSh 4,500 from her mandazi stall and mapped all her known costs (flour, charcoal, stall rent) against her earnings. They observed that some costs (rent) stayed the same regardless of how many mandazi she sold, while others (flour, charcoal) rose and fell with output. The class co-constructed a simple two-column income-vs-expenditure table on the board, making the deficit or surplus immediately visible.	A budget is a formal mathematical plan that compares total income with total expenditure over a defined period. Costs are classified as fixed (unchanging regardless of activity level — e.g., stall rent, club membership fees) or variable (changing with the level of activity — e.g., cost of flour per batch). When expenditure exceeds income the budget is in deficit; when income exceeds expenditure it is in surplus. Learners also established the vocabulary anchor for the entire sub-strand: every subsequent lesson is one tool Amina can use to improve her budget position.	The budget surplus or deficit is calculated as: Surplus/Deficit = Total Income − Total Expenditure. A positive result is a surplus; a negative result is a deficit. Pedagogical note: Emphasise that the formula is directional — income minus expenditure, not the reverse. Teachers should cold-call at least three students to classify a given cost as fixed or variable before moving on, allowing 10–15 seconds of wait time per student. Connect back to the driving question by asking: 'Which part of Amina's budget can commercial arithmetic help her improve?' — this seeds curiosity for Lessons 2–7.
Lesson 2 **Discounts and Percentage Discounts — How Much Does Amina Really Save at the Kondele Electronics Stall?**	Students viewed a scenario in which a trader at Kondele Market, Kisumu, advertised a blender at a marked price of KSh 3,200 with a '15% off' sticker. Learners first predicted the sale price, then worked through the arithmetic to verify. They observed that the shilling saving (discount) and the percentage saving (percentage discount) convey different but complementary information — the shilling	Discount is the actual shilling reduction on the marked price: Discount = Marked Price − Sale Price. Percentage Discount expresses that reduction as a proportion of the marked price: Percentage Discount = (Discount ÷ Marked Price) × 100. Both formulas are reversible — given a percentage discount, students can recover the sale price or the marked price. Learners also recognised that a larger percentage	The two core formulas are: (1) Discount = Marked Price − Sale Price; (2) Percentage Discount = (Discount ÷ Marked Price) × 100. A useful single-step sale price formula is: Sale Price = Marked Price × (1 − Percentage Discount ÷ 100). Pedagogical note: Teachers should present at least one problem in each direction (find discount given prices; find sale price given percentage discount) before independent

	amount tells Amina what stays in her pocket; the percentage allows fair comparison across items of different prices.	discount is not always a larger shilling saving if the marked prices differ.	practice. Use Kenyan market contexts — e.g., a mitumba stall in Gikomba offering 'KSh 200 off' vs. a supermarket offering '10% off' on different-priced items — to highlight why percentage comparison matters. Reconnect to the driving question: a savvy Amina always converts a shilling discount into a percentage before comparing offers.
Lesson 3 **Discounts and Percentage Discounts (Part 2 / Practice) — Which Deal Gives Amina Better Value?**	Students worked through a structured set of multi-step problems in which Amina compares three competing offers for the same cooking equipment from different stalls — one quoting a shilling discount, one quoting a percentage discount, and one offering a 'buy two, get one free' deal that must be converted into an equivalent percentage discount. Learners observed that surface-level advertising language can obscure the true saving and that a common percentage basis enables fair comparison.	All three discount formulas — Discount = Marked Price − Sale Price; Percentage Discount = (Discount ÷ Marked Price) × 100; Sale Price = Marked Price × (1 − Percentage Discount ÷ 100) — can be applied in reverse to find any unknown when two of the three quantities are known. Learners practised choosing the correct formula entry point depending on what information is given, a key problem-solving skill. They also learned that 'best value' requires a common unit of comparison (percentage discount), not just the largest shilling figure.	Pedagogical note: This is a deliberate consolidation and transfer lesson. Teachers should resist re-teaching Lesson 2 content; instead, use Socratic questioning — 'What do you know? What do you need? Which formula connects them?' — and allow a minimum of 2 minutes of independent think time before pair discussion. Cold-call at least four students to justify their formula choice, not just their answer. The lesson's link to the driving question is explicit: Amina uses percentage discount as a decision-making tool, not just a calculation, to protect her KSh 4,500 budget.
Lesson 4 **Commission and Percentage Commission — Who Gets Paid When Amina Sends Money to Uganda?**	Students examined a realistic M-Pesa / forex-bureau scenario in which Amina sends KSh 10,000 to a supplier in Kampala. The agent charges a commission. Learners observed the transaction receipt showing the principal amount, the commission deducted, and the net amount received. They also examined a real-estate agent scenario in which a Nairobi agent earns 3% commission on a KSh 2.4 million property sale — making the concept of commission as an incentive for facilitating transactions concrete.	Commission is the fee paid to an agent or intermediary for facilitating a financial transaction. It is calculated as: Commission = (Commission Rate ÷ 100) × Principal Amount. Percentage Commission = (Commission ÷ Principal Amount) × 100. Commission is always calculated on the principal (the transaction value), not on the agent's cost — a distinction that prevents a common student error. Learners also understood commission as both a cost to Amina (when she is the sender/buyer) and a potential income stream (if she herself acts as an agent).	The two core formulas are: (1) Commission = (Rate ÷ 100) × Principal; (2) Percentage Commission = (Commission ÷ Principal) × 100. Pedagogical note: Highlight the structural parallel with percentage discount — both express a portion of a reference amount as a percentage — so students can leverage prior knowledge. Use M-Pesa tariff tables (familiar to all Kenyan learners) as a real data source; have students calculate commission for three different transaction amounts and observe whether the percentage stays constant even as the shilling amount changes (it does for a fixed-rate commission, reinforcing proportional reasoning). Reconnect to the driving question: Amina must account for commission as a budget line item, not an afterthought.
Lesson 5 **Profit and Loss (Part 1 — Concepts & Calculation)**	Students traced Amina's journey from Wakulima Market in Nairobi, where she buys maize flour in bulk at KSh 1,800 per	Profit arises when SP > CP: Profit = SP − CP. Loss arises when CP > SP: Loss = CP − SP. Percentage Profit = (Profit ÷ CP) ×	The four formulas are: Profit = SP − CP; Loss = CP − SP; % Profit = (Profit ÷ CP) × 100; % Loss = (Loss ÷ CP) × 100.

— From Wakulima Market to Amina's Stall	10 kg bag, to her stall where she sells mandazi at retail prices. They observed that cost price (CP) and selling price (SP) are distinct concepts, and that the gap between them is not automatically profit — overheads have not yet been included (that complexity is saved for Lesson 6). The class also observed a loss scenario: a trader who bought sukuma wiki before a public holiday and was forced to sell at a reduced price before it wilted.	100; Percentage Loss = (Loss ÷ CP) × 100. Crucially, the percentage is always calculated on the cost price, not the selling price — this is the most common examination error and must be explicitly flagged. Learners also practised working backwards: given a percentage profit and CP, find SP; given SP and percentage loss, find CP.	Pedagogical note: Write 'ALWAYS divide by CP' prominently and return to it at the start of every worked example. Use a colour-coded number line (CP on the left, SP to the right for profit, SP to the left for loss) as a visual anchor. Teachers should present at least two worked examples — one profit, one loss — before independent practice. Cold-call students to state whether a transaction is profit or loss before they calculate, building conceptual clarity before procedural fluency. Connect to the driving question: understanding profit and loss is what separates a business that sustains Amina's income from one that quietly drains it.
Lesson 6 **Profit and Loss (Part 2 — Application & Problem Solving) — From Nyama Choma to Amina's Mandazi Stall**	Students analysed a detailed case study of a nyama choma trader in Kericho who purchased goat meat, paid for charcoal, transport from the Rift Valley, and stall hire — then sold at a set price. They observed that when only the purchase price of meat was treated as CP, the apparent profit was large; when all overheads were included in CP, the real profit shrank considerably and in one scenario became a loss. This created productive cognitive conflict that drove the lesson's key conceptual development.	Total Cost Price = Purchase Price of Goods + All Overheads (transport, packaging, labour, rent, fuel, etc.). Percentage profit and percentage loss must always be calculated on the TOTAL cost price, not just the purchase price. Learners also practised reverse problems: given selling price and percentage profit, recover total cost price; given total cost price and a desired percentage profit, calculate the required selling price. This makes the mathematics directly actionable for Amina's pricing decisions.	The key formula extension is: Total CP = Purchase Price + Sum of All Overhead Costs. All subsequent profit/loss calculations use this Total CP as the denominator. Pedagogical note: This lesson is the most common source of marks lost in KCSE-style assessments. Teachers should devote explicit time to listing overheads before any calculation begins — model this habit by always writing 'Step 0: List all costs' on the board. Use a think-pair-share structure for at least one problem: students individually list costs, compare with a partner, then the class constructs a consensus list before calculating. Reconnect to the driving question: Amina cannot make a smart money decision if she ignores hidden costs — this lesson makes her financially literate.
Lesson 7 **Currency Conversion — Crossing Borders with Amina's Shillings**	Students examined a realistic forex bureau board (as seen at Jomo Kenyatta International Airport or Westlands bureaux) showing buying and selling rates for USD, UGX, TZS, and EUR against KES. They observed that the bureau always buys foreign currency at a lower rate and sells it at a higher rate — creating a spread that is the bureau's profit. Amina's cross-border payment scenario (sending money to a Ugandan supplier and receiving payment	To convert KES into foreign currency, divide by the selling rate (the rate at which the bureau sells foreign currency to the customer). To convert foreign currency into KES, multiply by the buying rate (the rate at which the bureau buys foreign currency from the customer). The spread (Selling Rate − Buying Rate) represents the bureau's commission per unit of foreign currency. Learners practised both directions and calculated the total KES cost of a	The two conversion rules are: (1) Foreign Currency Amount = KES Amount ÷ Selling Rate; (2) KES Amount = Foreign Currency Amount × Buying Rate. Pedagogical note: Students almost universally confuse which rate to use in which direction. A reliable mnemonic: 'The bureau always wins' — when you want foreign currency, use the higher (selling) rate, which costs you more KES; when you return foreign currency, use the lower (buying) rate, which gives you

	from a Tanzanian customer) made both conversion directions immediately relevant.	transaction including the spread.	fewer KES. Reinforce with at least three direction-switching problems. Cold-call students to state the rule before they reach for a calculator. Connect to the driving question: Amina's informed use of conversion rates prevents her from losing money at the border — a real and immediate concern for traders near Busia, Namanga, or Lunga Lunga.
Lesson 8 **Synthesis, Model Revision & Final Explanation — Amina's Complete Financial Plan**	Students revisited the full Amina scenario introduced in Lesson 1 — now enriched with six weeks of learning. They observed their own Lesson-1 budget predictions alongside the complete model built across the sub-strand, noting where their initial assumptions were wrong (e.g., ignoring commission costs, not accounting for overheads in profit calculations, misjudging the effect of currency spreads). The contrast between the naive Lesson-1 model and the refined Lesson-8 model made learning growth tangible and visible.	All six commercial arithmetic tools — (1) budget preparation (income vs. expenditure, fixed vs. variable costs), (2) discount and percentage discount, (3) commission and percentage commission, (4) profit and loss on basic transactions, (5) profit and loss including overheads, and (6) currency conversion using buying and selling rates — are interconnected decision-making instruments. A complete financial plan requires all six to be applied simultaneously and coherently. Learners consolidated the understanding that arithmetic is not an abstract school exercise but a protective and empowering tool for every Kenyan who participates in markets, clubs, or cross-border trade.	Pedagogical note: This lesson should be structured as a guided model-revision session, not a new-content lecture. Teachers should project the class's Driving Question Board (accumulated across all 8 lessons) and systematically answer each posted question using the sub-strand's six tools. Allocate at least 15 minutes to student-led explanation — cold-call groups to present one tool each, then show how it modifies Amina's budget outcome. End with a written exit task: 'Write one smart money decision Amina should make, and show the arithmetic that supports it.' This closes the loop on the driving question and serves as a formative assessment of transfer. Remind students that the same six tools apply to their own household budgets, school club accounts, and future business ventures across Kenya.